

# Project Info & WRS

## GuideMate

Team Website: <https://nickhegde.github.io/guidemate/>

Submitted for: **SE 6361.001**

Phase I – First Report

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# 1. Introduction

Our current assignment involves the software engineering of a mobile application designed to assist individuals with visual impairments in navigating indoor spaces. More specifically, the app's functionality can be tailored to cover areas as limited as a specific section on one floor within the ECSS, ECSN, or ECSW premises. Our primary emphasis in the upcoming phases will be on the comprehensive process of Requirements Engineering to ensure the app aligns seamlessly with the unique needs and constraints of these specified environments

## 1.1 Purpose

The primary objectives of GuideMate are:

- Providing **voice-based navigation** for visually impaired users.
- Detecting obstacles and issuing **real-time alerts**.
- Offering **emergency assistance** by notifying pre-registered contacts.

## 1.2 Scope

GuideMate is designed to be used **indoors**, specifically in structured environments such as **universities, shopping malls, hospitals, and airports**. It relies on a combination of **computer vision, speech recognition, and real-time processing** to aid users.

Key features include:

- **Voice-command navigation** for hands-free interaction.
- **Real-time obstacle detection** using AI-powered image processing.
- **Emergency response system** for critical situations.
- **Personalization settings**, including adjustable navigation speed and preferred routes.

## 2. Issues:

Issues with the Preliminary Definition Give (ambiguities, incompleteness, inconsistency, conflicts, ...) as well as resulting discussions that occurred within our team.

### 2.1 Issues with II.1 The Domain, Stakeholders, Functional and Non-Functional Objectives

#### 2.1.1 Issue 1 - Stakeholders

Description: The stakeholders “By” (not included in PrelimGiven)

Options:

- Add those who are making, designing, or working on the app
- Add those who provide input/direction

Decision & Rationale: We added the first category but not the second category. Providing input in the form of a questionnaire is better suited for the ‘of’ category than the ‘by’ category.

Note: Programmers could be explicitly included as well to make sure they are considered when designing requirements.

#### 2.1.2 Issue 2 - Stakeholders

Description: The stakeholders “For” (updated from PrelimGiven)

Options:

- Add those who are making, designing, or working on the app
- Add those who provide input/direction
- Add those who are second-hand users of the app, or sometimes primary (first responders, helpers, etc.)

Decision & Rationale: We added the first category but not the second category. Providing input in the form of a questionnaire is better suited for the ‘of’ category than the ‘by’ category. The third category was discussed and since we recall it was to be added, it was added.

### **2.1.3 Issue 3 - Stakeholders**

Description: The stakeholders “Of” (not included in PrelimGiven)

Options:

- UTD is theoretically in charge of paying for this app
- We are in a class, and ownership goes to Lawrence Chung
- The team lead(s) are responsible for the app
- The app is not being funded, and will be owned by the users

Decision & Rationale: We have gone with the last three cases. The first is not true as UTD is not paying for this app. The second is true as we’re designing for a class led by Lawrence Chung and ultimately decided by Lawrence Chung. The third is true as these are the actual people who will likely own the repos and source code. The fourth is true as being free and open-sourced, the ‘boss’ is really whoever we are making this for.

### **2.1.4 Issue 4 - Domain**

Description: The environment is described strictly as the building

Options:

- The domain should include non-permanent obstacles
- The domain should include phone capabilities
- The domain should include services to connect to

Decision & Rationale: We have opted to include all of these in our domain. The latter two are less clearly in domain and may be in either the system or 3rd party system, but we considered they are still in the domain.

### **2.1.5 Issue 5 - Objectives**

Description: The objectives were not complete

Options:

- Expand the objectives description

Decision & Rationale: We expanded the objectives

## **2.2 Issues with II.2 Software System Functional Requirements**

### **2.2.1 Issue 1 - To E\_v or not to E\_v**

Description: Do the Software System FRs need to follow the “If E\_v then S\_v” format?

Options:

- Yes. He has said in class that requirements must be in the aforementioned form.
- No. His class statements referred to when using the Reference Model, which we are not using. Furthermore, the examples of corrected requirements in the document do not use this format.

Decision & Rationale: Option 2 was found more compelling. We also note that this is the software system requirements not the software system specifications and should be analogous to the “R” in our prior venn diagram, not the “S”

### **2.2.2 Issue 2 - “Shall”**

Description: The requirements given didn’t use “shall”

Options:

- We shall use “shall”
- We shan’t use “shall”

Decision & Rationale: Option 1 as “shan’t” is an odd old word, and we have no recollection of the professor saying it.

### **2.2.3 Issue 3 - IDs**

Description: The requirements weren’t uniquely identified

Options:

- Add full identifiers to ours
- Add sub identifiers to ours

Decision & Rationale: Option 1 would give bullets: 3.2.1.4 while Option 2 would give section 3.2.1 a list of 1 and 2. We decided the latter is sufficient as it allows for references in an identical matter, but is far more readable in the requirements section.

## **2.3 Issues with II.3 Software System Non-Functional Requirements**

### **2.3.1 Issue 1 - FRs from NFRs**

Description: We are to operationalize our NFRs into terms of FRs

Options:

- Put a list of FRs under every NFR showing what FRs each NFR changed into
- Have FRs in the FR section only

Decision & Rationale: We chose not to, as it prevents duplication as many were already FRs for other reasons, and it keeps the NFR list readable

### **2.3.2 Issue 2 - Conflicting NFRs**

Description: Some NFRs in the PrelimGiven are potentially at conflict, namely “fastest” route, and “route most most comfortable with”

Options:

- Reword the two such that they aren’t at odds
- Merge the two such that they are under a new unified NFR

Decision & Rationale: We can’t think of how to keep them separate and not at odds, so we merged them into a clearer single NFR

### **2.3.3 Issue 3 - Unclear NFRs**

Description: The system being “ubiquitous” is unclear

Options:

- Reword ubiquitous to be clearer
- Remove the requirement

Decision & Rationale: In no sense of the word is our app planning to be ubiquitous that we can think of, so we removed it.

## **2.4 Other Issues**

### **2.4.1 Issue 1 - Questionnaire 1**

Description: Should we have multiple questionnaires?

Options:

- Yes, as we can tailor our questions to the type of person being asked

- No as it makes it more streamlined and could reduce bias

Decision & Rationale: Option 2 won out

### 2.4.2 Issue 2 - Questionnaire 2

Description: How tailored should the questionnaire be?

Options:

- Specific questions for the sorts of people we want (blind, helpers, or first responders)
- Questions not as clearly tailored for those

Decision & Rationale: The former allows us to better understand what certain groups would like, while the latter allows us to better understand what the general populace would benefit from and keeps the purpose of the survey hidden, potentially reducing bias. We choose the former, but with most questions rather general.

## 3. WRS

The WRS is an improved description of the problem and highest level solution, largely adapted from the Preliminary Definition Given

a. W

i. Problem

Blind people can have difficulty navigating safely even when indoors. There can be changing obstacles such as other people, event tables, and spills. Even under ideal circumstances, there can be difficulties in knowing how to get to a destination since ECS signage (other than room numbers) is too high and has no braille.

(See slides for AS-IS to TO-BE scenarios)

ii. Goal

iii. We need to make an app that helps blind people navigate indoors at some UTD buildings *safely*.

iv. Improved understanding of II.1 The Domain, Stakeholders, Functional and Non-Functional Objectives

Domain

The domain of BlindWalker is to help visually impaired individuals navigate indoors. The indoor premise will be restricted to the entirety of the first floor of ECSW, ECSS, or ECSN. The primary focus of our app is to allow users to navigate to their destination safely, while characteristics such as the duration and convenience of the route are other metrics we are keeping in mind. This will require app functionalities such as object detection, accurately calculating relative distance and movement, and presenting all this information to the user in an intuitive, concise and non-visual way, among many other necessary services. Alongside this, when designing our app, we are assuming the users have a basic understanding of the usage of their device and are familiar with the features available from their smartphone.

As mentioned before, a key challenge of this domain is the safety of our users.

Considering this, key actors apart from the BlindWalker system include the University Police, First Responders, and Medical Personnel in the case of an emergency or accident during navigation.

In order to achieve our primary objectives, we will need key information from the users such as their severity of impairment or if they already use accommodations such as a cane or

service dog. Alongside this, the application must store data about the navigation such as a comprehensive log of the events that occurred while on route.

## Stakeholders

One of the most important parts to any software project are the stakeholders. Stakeholders of a project are important, and they include the people working on the project, the people the project is made for, and the people who are going to use the software project. In our case, we can divide these into 3 types: “Of the People”, “By the People”, and “For the People”.

“Of the People” corresponds to the people whose vision of the product we want to accomplish. For our system, our project leads are Logan Dane and Joseph Farghal, who are responsible for overseeing the requirements process and ensuring the quality of the requirements that we elicit. The main requirements were given out by Dr. Chung. We also created a questionnaire to interview blind people to gather more requirements for our system. Some of the questions we asked were about their current challenges, preferences, and expectations for an indoor navigation system.

“By the People” matches with the people that are actually going to be building the system. These include the requirements engineers, who analyze the needs and specifications of the system and elicit more requirements from the stakeholders, the software design architects, who design the structure and components of the system, system programmers, UI/UX designers, and the test engineers. Who verify and validate the functionality and performance of the system. Our requirements team is composed of 6 members who are all working to elicit requirements. “For the People” refers to the people that we are developing the system for. These are the end-users who will benefit from the software project. In our case, these are the blind and visually impaired people who need a way to navigate indoors safely. Blind people will not be the only people using the system, however. There are those who assist blind people (such as their caretakers) as well as emergency services. The assistants would be able to work together with the system to navigate the blind even more safely, while the emergency services could use it to review the last moments before an accident that injured the blind user. We created the questionnaire in order to ask more questions for the potential users so that we know exactly what they would like out of the system.

## 3.1 Functional and Non-Functional Objectives

- 3.1.1** The main functional objective of Safe Trek is to significantly improve the accessibility of indoor navigation for the visually impaired, offering a navigation solution that is both seamless and precise. Safe Trek will introduce multiple methods for users to input their destinations, deliver clear and audible instructions for route guidance, and adjust navigation routes in real-time based on user preferences, including options to avoid stairs. The primary goal is to enhance the autonomy and confidence of visually impaired individuals, facilitating their ability to navigate complex spaces without assistance.
- 3.1.2** In addition to its functional goals, Safe Trek sets out to redefine excellence in safety, user-centric design, and system reliability. Safety is the cornerstone of Safe Trek’s design, with every feature meticulously crafted to minimize the risks inherent in indoor navigation for visually impaired users. The system prioritizes a user-friendly interface that can be extensively customized to cater to the unique needs and preferences of each user, ensuring the navigation solution is accessible to all. Engineered for consistent performance across a diverse range of environments and efficient in its battery usage,

Safe Trek aims for reliability in every use case. Further, it will also take standard measures to preserve data privacy.

## **3.2 Functional and Non-Functional Requirements**

### **3.2.1 Functional RS - Improved understanding of II.2 Software System Requirements: FRs**

1. The system shall allow the user to input a destination location prior to beginning navigation.
2. The system shall allow for user input in the form of typing or speaking.
3. The system shall accept user input in the form of a valid room number (such as ECSW 3.250) or a common name.
4. The system shall confirm, out loud, the name of the destination location prior to beginning navigation.
5. The system shall allow the user to modify the volume of the system's audible messages.
6. The system shall allow the user to terminate navigation at any time when navigation is in progress.
7. The system shall notify the user when the final destination has been reached via text to speech.
8. The system shall terminate navigation when the final destination has been reached.
9. The system shall allow the user to specify preferences prior to beginning navigation, such as avoiding elevators or stairs.
10. The system shall calculate multiple routes to reach the destination which respect the user's preferences, so that the user has different options to choose from.
11. The system shall allow the user to save a specific route and destination for easy access.
12. The system shall allow the user to, instead of inputting a destination prior to beginning navigation, simply select a saved route.
13. The system shall display and periodically verbalize the approximate time and distance to the next location.
14. The system shall provide instructions as short, imperative sentences with all distances explicitly quantified, such as "Stop" or "Turn right in 3 steps".
15. The system shall allow the user to customize the time interval between these verbalizations.
16. The system shall allow the user to customize the unit of measurement used for the distance, such as number of steps or meters.
17. The system shall prompt the user to allow camera access permissions to the app, if not already allowed.
18. The system shall prompt the user to hold their mobile device upright so that the camera is facing forward.
19. The system shall utilize the device's built-in camera to visually detect obstacles.
20. The system shall warn the user if the camera's lenses become obstructed.
21. The system shall verbally warn about the distance and location of obstacles (including objects and people) well in advance.
22. The system shall provide concise spoken instructions to avoid collision with said obstacles.
23. The system shall attempt to detect falls using the camera and/or built-in sensors.
24. The system shall prompt the user whether to place emergency calls and messages upon detecting a fall or being unable to figure out the current location, which gives the user a chance to cancel the emergency actions if unnecessary (such as if the system mistakes dropping the phone for a fall).
25. If the user agrees to place the emergency calls or remains unresponsive when prompted, the system shall proceed to place emergency calls and messages to the emergency contact.
26. The system shall prompt the user to save an emergency contact (a phone number and optionally a name) before accepting input destinations, if an emergency contact has not yet been saved.
27. The system shall warn the user before or during navigation when battery power is low (below 20%).

### **3.2.2 Non-functional RS - Improved Understanding of II.2 Software System Requirements: NFRs**

1. The system shall help the user safely navigate indoors.
2. The system shall lead the user through the fastest route which still matches the user's navigation preferences.



3. The system shall be usable.
4. The system shall work within all areas of ECSW, ECSS, or ECSN.
5. The system shall be customizable to every user (e.g. speaking volume, speaking rate, speaking intervals, units of measurement, etc.).
6. The system shall be extensible to accommodate variations such as different interfaces, operating systems, system language, new features, new sensors and hardware, etc.
7. The system shall provide clear and understandable instructions.
8. The system shall be maintainable to accommodate changes to floor plans, out-of-service elevators, off-limits areas, etc.
9. The system shall securely store or purge the user's current location and emergency contacts as per standard data privacy regulations.
10. The system shall perform efficiently while in use.
11. The system shall have reasonably low battery consumption to allow for prolonged usage.
12. The system shall be compatible with screen readers.
13. The system shall make basic features available even when offline.

### **3.3 Traceability**

| Typ<br>e | Requireme<br>nt # | World                                                                                                   | Explanation                                                                                                                   |
|----------|-------------------|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| FR       | 1                 | Blind People need help navigating safely indoors                                                        | This requirement lets blind people select destinations to travel to.                                                          |
| FR       | 2                 | Blind people cannot see the phone screen                                                                | This requirement lets blind users interact with the system.                                                                   |
| FR       | 3                 | ECSW has names for locations to use.                                                                    | This requirement allows precise navigation of ECSW.                                                                           |
| FR       | 4                 | Blind people cannot see visual feedback from the app                                                    | This requirement lets blind people get audio feedback from app.                                                               |
| FR       | 5                 | Sometimes audio is drowned out by other noises in the area                                              | This requirement allows the blind person to change the volume as needed.                                                      |
| FR       | 6                 | Sometimes, people realize they want to go somewhere else.                                               | This requirement allows the user to change their desired destination.                                                         |
| FR       | 7                 | Blind people cannot see if they have arrived at a destination.                                          | This requirement allows blind people to know when they make it to their destination.                                          |
| FR       | 8                 | After reaching the destination, people do not need to go anywhere else.                                 | This requirement stops the navigation after reaching a destination.                                                           |
| FR       | 9                 | Blind people have different preferences, based on taking stairs or the elevator.                        | This requirement allows the user to decide how they want to move between floors.                                              |
| FR       | 10                | There are many different paths to take to reach a destination                                           | This requirement allows the user to choose one of multiple paths to account for the many different paths that could be taken. |
| FR       | 11                | People can have favorite routes that they would like to take.                                           | This requirement lets users save their favorite routes.                                                                       |
| FR       | 12                | People can have favorite routes that they would like to take.                                           | This requirement lets users easily use their favorite routes.                                                                 |
| FR       | 13                | Blind people do not know how long it takes to walk somewhere, and cannot see what their phone displays. | This requirement has the system vocalize the time and distance left in their travel.                                          |
| FR       | 14                | Blind people need fast communication to have time to react to obstacles.                                | This requirement makes it so the system vocalizes short and precise directions so that blind people have time to react.       |
| FR       | 15                | Too many or too few audible commands can be problematic to users.                                       | This requirement lets the user specify how often to provide directions.                                                       |
| FR       | 16                | The preferred and familiar unit of measurement is likely to vary between users.                         | This requirement allows the user to specify what units to use                                                                 |
| FR       | 17                | The user should not be forced to use a camera, and should be aware when one is used.                    | This requirement asks the user for permission before using a camera                                                           |
| FR       | 18                | Blind people may not be aware in what way they are holding something.                                   | This requirement makes sure the system helps the phone be oriented correctly.                                                 |
| FR       | 19                | Blind People need help navigating safely indoors                                                        | This requirement gives the phone sight to help the blind person                                                               |
| FR       | 20                | Blind people may not be aware in what way they are holding something.                                   | This requirement keeps the phone's sight and lets the user know if they are covering.                                         |

|         |    |                                                                           |                                                                                                                               |
|---------|----|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| FR      | 21 | Blind people need fast communication to have time to react to obstacles.  | This requirement makes the system provide queues early enough to be useful.                                                   |
| FR      | 22 | Blind people need fast communication to have time to react to obstacles.  | This requirement makes the system provide clear enough queues to be useful.                                                   |
| FR      | 23 | Blind people may fall and come to harm.                                   | This requirement make the system aware if the user has fallen (or seems like it)                                              |
| FR      | 24 | Blind people may or may not want emergency services if they fall.         | This requirement gives the user a choice to get help or not.                                                                  |
| FR      | 25 | Blind people may or may not want emergency services if they fall.         | This requirement helps the user get emergency services.                                                                       |
| FR      | 26 | Blind people may want to contact specific people if an emergency happens. | This requirement helps the user get a emergency contact before they are needed.                                               |
| FR      | 27 | Blind people may be stranded if their phone dies while out.               | This requirement helps the user know that their phone is low battery when it is.                                              |
| NF<br>R | 1  | Blind People need help navigating safely indoors                          | This requirement emphasizes the importance of safety.                                                                         |
| NF<br>R | 2  | Blind People need help navigating safely indoors                          | This requirement emphasizes the implicit efficiency of the route.                                                             |
| NF<br>R | 3  | Blind people cannot see the phone screen                                  | This requirement emphasizes the useability of the app for blind people.                                                       |
| NF<br>R | 4  | Blind People need help navigating safely indoors                          | This requirement specifies where the navigation is to occur                                                                   |
| NF<br>R | 5  | Blind people have different preferences                                   | This requirement emphasizes the need for customization                                                                        |
| NF<br>R | 6  | Blind people have different phone systems.                                | This requirement allows many different systems to run our app.                                                                |
| NF<br>R | 7  | Blind people need fast communication to have time to react to obstacles.  | This requirement makes it so that blind people have time to react to the instructions since they are simple and clear.        |
| NF<br>R | 8  | Buildings are often under construction and need changes.                  | This requirement allows the system to update to accommodate changes in the building.                                          |
| NF<br>R | 9  | Keeping information of user's secure is important                         | This requirement makes the system secure and keeps their data private.                                                        |
| NF<br>R | 10 | Blind People need help navigating safely indoors                          | This requirement keeps the system running at real time so that users do not have to wait for instructions and miss obstacles. |
| NF<br>R | 11 | Blind people need prolonged help navigating safely indoors                | This requirement keeps the system optimized so that it can help people longer.                                                |
| NF<br>R | 12 | Blind people have different preferences for getting feedback              | The system shall be compatible with screen readers so that blind people can use their own personal setup.                     |
| NF<br>R | 13 | Blind people may not always have access to the internet.                  | This requirement makes it so that blind people can get help navigating safely indoors even without internet connection.       |

## 4. System Architecture and Design

### 4.1 Tools and Technologies

| Component     | Technology                           |
|---------------|--------------------------------------|
| Design        | Figma                                |
| Frontend      | TensorFlow.js for obstacle detection |
| AI & ML       | TensorFlow.js for obstacle detection |
| Accessibility | Google Text-to-Speech API            |

### 4.2 Workflow

1. **User inputs a destination** using voice or text.
2. **GuideMate processes the request** and determines the best route.
3. **Obstacle detection** runs continuously, warning the user in real-time.
4. **Emergency detection** is active; saying 'EMERGENCY' triggers a call.

## 5. User Scenarios (AS-IS → TO-BE)

### 5.1 Scenario 1: User Navigation

#### AS-IS:

A visually impaired person struggles to navigate through a shopping mall without assistance.

#### TO-BE:

The user activates **GuideMate**, states their destination, and receives real-time voice guidance.

### 5.2 Scenario 2: Emergency Handling

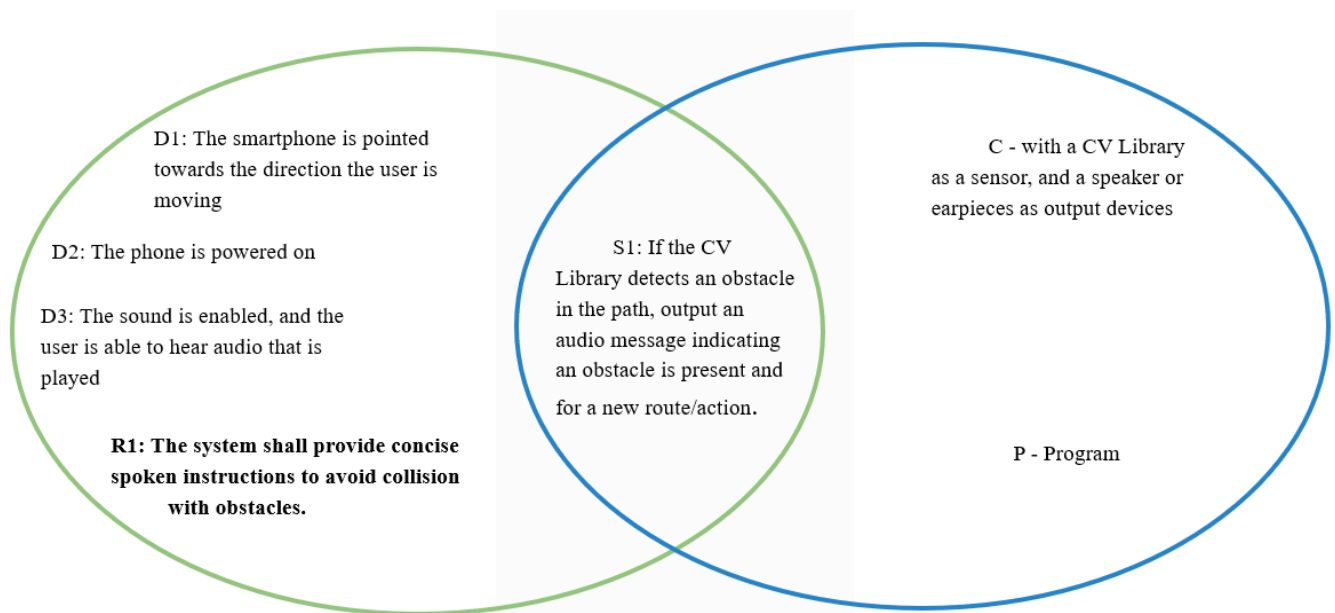
## AS-IS:

A visually impaired individual falls and has no immediate way to contact help.

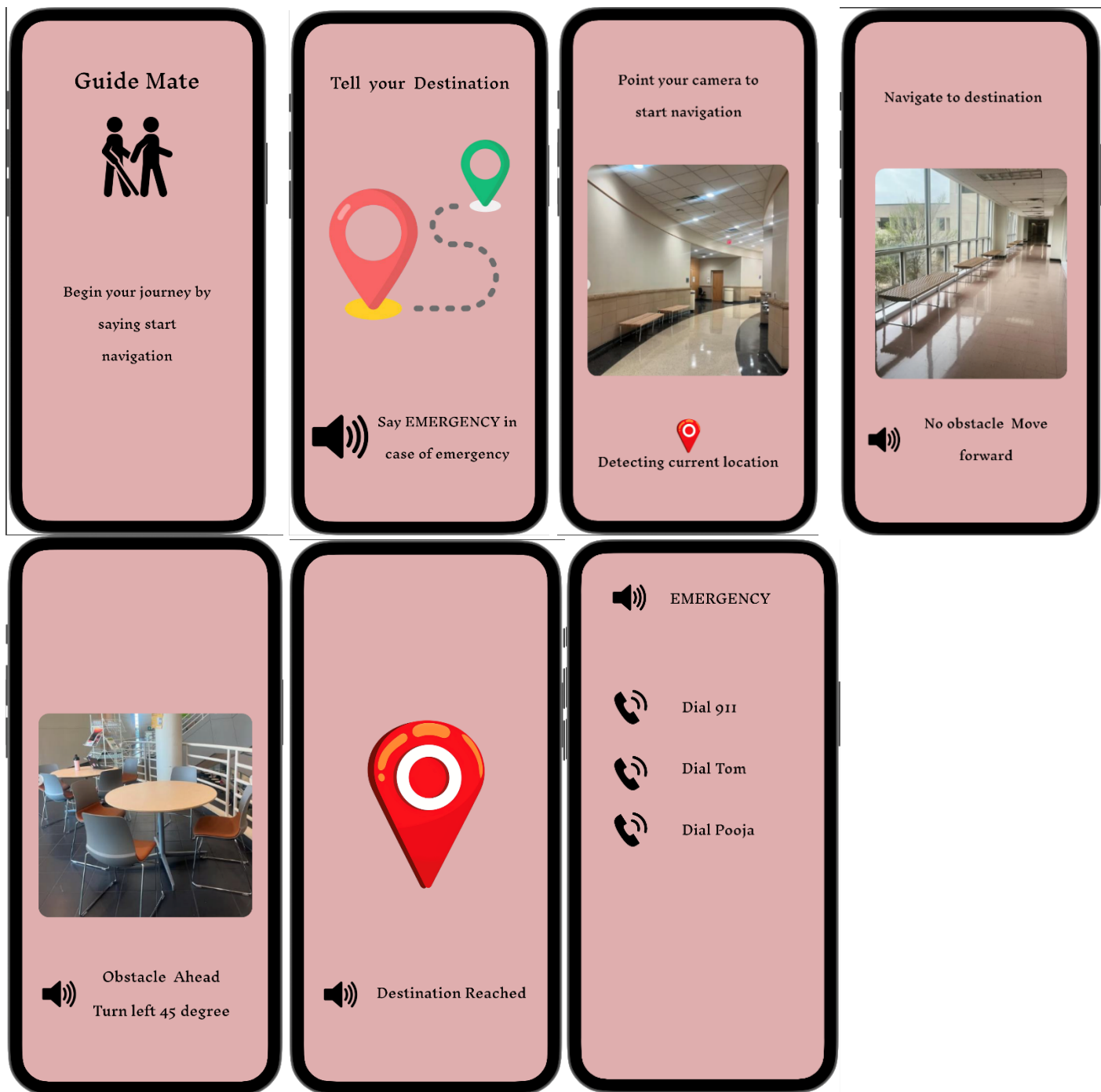
## TO-BE:

**GuideMate detects the emergency** and allows the user to say 'EMERGENCY' to call for help.

## 6. Venn Diagram



## 7. Results:



## 8. Conclusion & Future Work

GuideMate is designed to be a practical and efficient indoor navigation solution for visually impaired individuals. The system integrates critical dependencies such as proper smartphone orientation, power availability, and enabled sound settings to ensure smooth operation. It leverages a Computer Vision (CV) library as a sensor, coupled with audio output devices, to detect obstacles and provide real-time spoken instructions. By dynamically adjusting routes upon obstacle detection, GuideMate enhances safety and user experience.

Future work includes:

- Enhancing AI-driven obstacle detection for more accurate and timely alerts.
- Expanding integration with smart home devices for a seamless navigation experience.
- Implementing a crowdsourced real-time mapping system to improve indoor navigation accuracy.

GuideMate represents an optimized approach based on insights from previous projects and system design considerations, ensuring smarter, safer, and more efficient navigation for users.